A LOCAL SINGLE-CORE CRITERION FOR TYPE II NAVIER–STOKES BLOWUP

Abstract. We prove a local compact single-core exclusion criterion for Type II Navier–Stokes concentration. The theorem is formulated on fixed compact parabolic cylinders with explicit Caffarelli–Kohn–Nirenberg bounds, positive core concentration, localized scale-translation representation, local pressure reconstruction, and vanishing localized dissipation on selected windows. Compactness and pressure stability then force the selected limit to be spatially constant. A zero constant contradicts the positive CKN core; a nonzero constant gives a bounded selected-window limit, which is excluded by the definition of the local Type II regime.

1. Introduction

This paper isolates a local compactness mechanism for Type II concentration in the three-dimensional Navier–Stokes equations. The setting is a sequence of renormalized solutions on fixed parabolic cylinders, centered at a selected concentration core and written in scale-translation variables. The result excludes the case in which one compact core carries a fixed positive Caffarelli–Kohn–Nirenberg density while the localized dissipation along selected windows vanishes. The theorem is conditional on the local representation and pressure decomposition/stability inputs stated below; the compact-cylinder A, E, C, D bounds are part of the branch hypothesis.

The argument is local. Uniform Caffarelli–Kohn–Nirenberg bounds and bounded modulation coefficients give compactness on smaller cylinders. Vanishing localized dissipation forces the compactness limit to be spatially constant. If the constant is zero, strong velocity convergence and pressure stability contradict the positive core density. If the constant is nonzero, the sequence has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition used here.

2. Setup and definitions

For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R, define

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v(y,s)|^2 \, dy,$$

$$E_v(R) = R^{-1} \int_{Q_R} |\nabla v|^2 \, dy \, ds,$$

$$C_v(R) = R^{-2} \int_{Q_R} |v|^3 \, dy \, ds,$$

$$D_q(R) = R^{-2} \int_{-R^2}^{0} \int_{B_R} |q(y,s) - (q)_{B_R}(s)|^{3/2} \, dy \, ds.$$

All pressure quantities are understood modulo functions of time.

Definition 2.1 (Bounded selected-window limit)**.** Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r. A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \to V_* \qquad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r.

Definition 2.2 (Local Type II sequence)**.** A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of Theorem 2.1.

Definition 2.3 (Compact single-core vanishing-dissipation branch)**.** Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*}. The sequence is a compact single-core vanishing-dissipation branch if:

(i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation

$$\partial_s V_n + (V_n \cdot \nabla)V_n + \nabla P_n = \nu\Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \qquad \nabla \cdot V_n = 0; \tag{1}$$

(ii) the compact-cylinder suitable package is uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(iii) the selected core has positive CKN concentration on Q_{r_*}:

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0 \qquad \text{for all } n;$$

(iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

(v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M;$$

(vi) the branch has vanishing localized dissipation in the sense of Theorem 2.4 below.

Definition 2.4 (Vanishing localized dissipation)**.** Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \to 0.$$

3. Local analytic inputs

The proof uses the following local analytic statements. They are stated here in the form needed for the compact single-core argument.

Assumption 3.1 (Local representation)**.** For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.

Assumption 3.2 (Local pressure decomposition and stability)**.** For every $B_r \Subset B_R$, the represented pressure decomposes as

$$P_n = P_{\mathrm{loc},n} + H_n, \qquad -\Delta P_{\mathrm{loc},n} = \partial_i\partial_j(\zeta V_{n,i}V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r, and H_n is harmonic in B_r. The local part satisfies the Calderon–Zygmund bound

$$\|P_{\mathrm{loc},n}\|_{L^{3/2}(B_R)} \le C\|V_n\|^2_{L^3(B_R)}$$

for a.e. time, and the harmonic part is controlled modulo spatial means by the local pressure norm.

Moreover, if $V_n \to 0$ strongly in $L^3(Q_R)$, then for every $r < R$,

$$P_n - (P_n)_{B_r} \to 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

4. Compactness and pressure stability

Proposition 4.1 (Compactness on selected windows)**.** *Let (V_n, P_n) be suitable on Q_{R_*}, solve (1), and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V,P) such that

$$V_n \overset{*}{\rightharpoonup} V \quad \text{in } L^\infty(-r^2,0;L^2(B_r)),$$

$$V_n \rightharpoonup V \quad \text{in } L^2(-r^2,0;H^1(B_r)),$$

$$V_n \to V \quad \text{strongly in } L^3(Q_r),$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak- in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.*

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in L^2H^1 on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2,0;H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ, integrating by parts in space where needed, and using the embedding above gives

$$|\langle (V_n\cdot\nabla)V_n,\varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}V_{n,j}\partial_j\varphi_i\,dy\right| \le C\|V_n\|^2_{L^3(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle \nu\Delta V_n,\varphi\rangle| \le C\|\nabla V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle \nabla P_n,\varphi\rangle| = \left|\int_{B_{r_1}} (P_n-(P_n)_{B_{r_1}})\nabla\cdot\varphi\,dy\right| \le C\|P_n-(P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})}\|\varphi\|_{H^m}.$$

The modulation terms obey

$$|\langle a_nV_n,\varphi\rangle| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle a_n\, y\cdot\nabla V_n,\varphi\rangle| = \left|\int_{B_{r_1}} a_n V_{n,i}\partial_j(y_j\varphi_i)\,dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

and

$$|\langle b_n\cdot\nabla V_n,\varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}b_{n,j}\partial_j\varphi_i\,dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n. Integrating these estimates in time gives a uniform bound because

$$\int_{-r_1^2}^0 \|V_n(s)\|^2_{L^3(B_{r_1})}\,ds \le r_1^{2/3}\|V_n\|^2_{L^3(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})}\,ds \le r_1\|\nabla V_n\|_{L^2(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|P_n(s)-(P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})}\,ds \le r_1^{2/3}\|P_n-(P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})},$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})}\,ds \le r_1^2\|V_n\|_{L^\infty(-r_1^2,0;L^2(B_{r_1}))}.$$

The uniform C, E, D, and A bounds therefore prove the claimed time-derivative bound.

The compact embedding $H^1(B_{r_1}) \Subset L^2(B_r)$, together with Aubin–Lions, gives

$$V_n \to V \qquad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \le C(r,M).$$

Indeed, for a.e. s, the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \le C\|V_n(s)\|^{2/5}_{L^2(B_r)}\|\nabla V_n(s)\|^{3/5}_{L^2(B_r)}.$$

Raising to the power 10/3, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^0 \|V_n(s)\|^{10/3}_{L^{10/3}(B_r)}\,ds \le C\left(\operatorname*{ess\,sup}_{-r^2<s<0}\|V_n(s)\|^2_{L^2(B_r)}\right)^{2/3}\int_{Q_r}|\nabla V_n|^2\,dy\,ds \le C(r,M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \le \|V_n - V\|^{1/6}_{L^2(Q_r)}\|V_n - V\|^{5/6}_{L^{10/3}(Q_r)}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \to V \qquad \text{strongly in } L^3(Q_r).$$

The pressure compactness is the weak compactness from the uniform D-bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \qquad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. The only point requiring comment is the modulation. For every compactly supported test field φ,

$$\int a_n(s)\,y\cdot\nabla V_n\cdot\varphi = -\int a_n(s)\,V_{n,i}\,\partial_j(y_j\varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j(y_j\varphi_i) = 3\varphi_i + y_j\partial_j\varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n: if $F_n \to F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞, then

$$\int a_nF_n - \int aF = \int a_n(F_n - F) + \int (a_n - a)F \to 0.$$

The term a_nV_n is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = -\int V_{n,i}\, b_{n,j}(s)\, \partial_j\varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n. These passages give the limiting represented equation in the sense of distributions. □

Remark 4.2 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n, they allow the modulation terms to pass to the distributional limit once $V_n \to V$ strongly in L^2_{loc}.

Lemma 4.3 (Stability of local pressure decomposition). *Let $V_n \to V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of Theorem 3.2. Then for every $r < R$,*

$$P_{\mathrm{loc},n} \to P_{\mathrm{loc}} \quad \textit{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\mathrm{loc}} = \partial_i\partial_j(\zeta V_iV_j)$. If $V \equiv 0$ on Q_R, then

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

Proof. Strong convergence in L^3 gives

$$V_{n,i}V_{n,j} \to V_iV_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)} \le \big(\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)}\big)\|V_n - V\|_{L^3(Q_R)}.$$

Calderon–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\mathrm{loc},n} - P_{\mathrm{loc}}\|_{L^{3/2}(Q_r)} \le C_{r,R}\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)}.$$

Assume now that $V \equiv 0$ on Q_R. Then $P_{\mathrm{loc}} = 0$. By the zero-limit stability part of Theorem 3.2,

$$P_n - (P_n)_{B_r} \to 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

□

Lemma 4.4 (Zero-dissipation limit). *If*

$$V_n \rightharpoonup V \quad \textit{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I \int_{B_\rho} |\nabla V_n|^2\, dy\, ds \to 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ.

Proof. Weak lower semicontinuity gives

$$\int_I \int_{B_\rho} |\nabla V|^2 \, dy \, ds \le \liminf_{n\to\infty} \int_I \int_{B_\rho} |\nabla V_n|^2 \, dy \, ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s. □

Lemma 4.5 (Spatially constant limits)**.** *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ, with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from Theorem 4.4. Since $V(\cdot, s) = c(s)$ on B_ρ, the local $L^\infty_s L^2_y$ bound gives

$$|c(s)|^2 |B_\rho| \le \operatorname*{ess\,sup}_{-\rho^2<s<0} \int_{B_\rho} |V(y, s)|^2 \, dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the strong L^3 convergence of Theorem 4.1 gives a subsequential limit to a nonzero bounded function on Q_ρ, which is precisely a nonzero bounded selected-window limit in the sense of Theorem 2.1. □

5. Contradiction lemmas

Lemma 5.1 (Persistence of positive core concentration)**.** *For a compact single-core branch in the sense of Theorem 2.3, there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0 \qquad \textit{for all } n.$$

Proof. This is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. □

Lemma 5.2 (Zero constant contradicts positive CKN concentration)**.** *If the compactness limit in Theorem 4.1 is identically zero on Q_{r_*}, then*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0.$$

Consequently this case contradicts Theorem 5.1.

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 \, dy \, ds = r_*^{-2} \|V_n\|^3_{L^3(Q_{r_*})} \to 0.$$

By Theorem 4.3, the normalized pressure converges strongly to zero in $L^{3/2}(Q_{r_*})$. Therefore

$$D_{P_n}(r_*) = r_*^{-2} \|P_n - (P_n)_{B_{r_*}}\|^{3/2}_{L^{3/2}(Q_{r_*})} \to 0.$$

Thus

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting the uniform lower bound

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0.$$

□

Lemma 5.3 (Nonzero constant contradicts the Type II regime)**.** *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By Theorem 4.5, the selected-window subsequence has a nonzero bounded selected-window limit. This contradicts Theorem 2.2. □

6. Main theorem

Theorem 6.1 (Local single-core Type II criterion)**.** *Assume the local analytic inputs in Theorems 3.1 and 3.2. A compact single-core vanishing-dissipation branch in the sense of Theorem 2.3 cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds \to 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ, and in particular on the selected core cylinder Q_{r_*}.

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting persistence of positive core concentration.

If the spatially constant limit is not identically zero on Q_{r_*}, the nonzero-limit lemma produces a nonzero bounded selected-window limit, contradicting the definition of a local Type II sequence. The two cases exhaust all spatially constant limits, so the branch cannot occur. □

Remark 6.2 (Minimal local inputs). The proof uses two local analytic inputs from the Type II analysis: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are not treated inside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.